

**Technical Data Sheet**

Date : 01.07.2025  
Product Name : Transformer Oil - TRANSOL GIX II  
Grade : IS : 335 : 2018 – Type – II ( Inhibited ) & RDSO Specification  
Customer :

Rev. No. 01

| Sr.No.                                           | Property                                                     | Test Method                                                          | Limits                                                          | Typical Values                               |
|--------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------|
| <b>A - Function :</b>                            |                                                              |                                                                      |                                                                 |                                              |
| i)                                               | Viscosity at 40 °C                                           | IS :1448 ( Part 25 )                                                 | 15 mm <sup>2</sup> /s, Max                                      | 8.4                                          |
| ii)                                              | Viscosity at 0 °C                                            | IS :1448 ( Part 25 )                                                 | 1800 mm <sup>2</sup> /s, Max                                    | 40.5                                         |
| iii)                                             | Viscosity at - 30 °C                                         | IS : 1448 ( Part 25 )                                                | -                                                               | NA                                           |
| iv)                                              | Viscosity at - 40 °C                                         | IS : 16084                                                           | -                                                               | NA                                           |
| v)                                               | Pour Point                                                   | IS : 1448 ( Part 10/Sec2 )                                           | - 10 °C, Max                                                    | - 33                                         |
| vi)                                              | Water Content                                                | IEC 60814                                                            | To be based on LCSET refer Table 1 (i)                          | 12                                           |
|                                                  |                                                              | IEC 60814                                                            | 30 mg/kg (for bulk supply), Max                                 | 14                                           |
| vii)                                             | Breakdown voltage                                            | IS : 6792                                                            | 40 mg/kg (for delivery in drums), Max                           | 40 - 60                                      |
|                                                  |                                                              | IS : 6792                                                            | 30 kV (as delivered), Min                                       | -                                            |
| viii)                                            | Density at 20 °C                                             | IS : 1448 ( Part 16 )                                                | 70 kV (After laboratory treatment), Min                         | 0.825                                        |
|                                                  | Density at 29.5 °C                                           | IS : 1448 ( Part 16 )                                                | 0.895 g/ml, Max                                                 | 0.820                                        |
| ix)                                              | DDF at 90 °C                                                 | IEC 60247 OR IS : 16086                                              | 0.005, Max                                                      | 0.00020                                      |
| x)                                               | Particle content                                             | IS:13236                                                             | No general requirement                                          | -                                            |
| <b>B - Refining / Stability</b>                  |                                                              |                                                                      |                                                                 |                                              |
| xi)                                              | Appearance                                                   | -                                                                    | Clear, free from sediment and suspended matter                  | Clear, free from sediment & suspended matter |
| xii)                                             | Acidity                                                      | IEC 62021 - 1                                                        | 0.01 mg KOH/g, Max                                              | 0.0015                                       |
| xiii)                                            | Interfacial Tension , mN/m                                   | ASTM D 971                                                           | No general requirement                                          | 47                                           |
| xiv)                                             | Total Sulphur Content                                        | ISO 14596 or ASTM D 4294                                             | No general requirement                                          | 0.002                                        |
| xv)                                              | Corrosive Sulphur                                            | DIN 51353                                                            | Not corrosive                                                   | Non corrosive                                |
| xvi)                                             | Potentially Corrosive Sulphur                                | IS: 16310                                                            | Not corrosive                                                   | Non corrosive                                |
| xvii)                                            | DBDS                                                         | IS: 16497 ( Part 1 )                                                 | Not detectable (< 5 mg/kg)                                      | Not detectable                               |
| xviii)                                           | Inhibitors according to IS : 13631/ IEC 60666                | IS: 13631                                                            | (I) Inhibited Oil; 0.25 - 0.30 (RDSO Spec)                      | 0.28                                         |
| xix)                                             | Metal passivator additives according to IS 13631 / IEC 60666 | IS: 13631                                                            | Not detectable (< 5 mg/kg) or as agreed upon with the purchaser | Not detectable                               |
| xx)                                              | Other additives                                              | -                                                                    | See 7                                                           | Not added                                    |
| xxi)                                             | 2 - Furfural & related compound content                      | IS : 15668                                                           | Not detectable (< 0.05mg/kg) for each individual compound       | Not detectable                               |
| <b>C - Performance :</b>                         |                                                              |                                                                      |                                                                 |                                              |
| xxii)                                            | Oxidation Stability                                          | IS 12422( Method C )<br>Test duration :<br>(I) Inhibited Oil : 500 h |                                                                 |                                              |
| a)                                               | Total acidity                                                | 1.9.4 of IS 12422                                                    | 0.2 mg KOH/g, Max ( RDSO Spec )                                 | 0.12                                         |
| b)                                               | Sludge                                                       | 1.9.1 of IS 12422                                                    | 0.05 %, Max ( RDSO Spec )                                       | 0.020                                        |
| c)                                               | DDF at 90 °C                                                 | 1.9.6 of is 12422                                                    | 0.5, Max                                                        | 0.035                                        |
| xxiii)                                           | Gassing tendency                                             | IEC 60628, Method A                                                  | No general requirement                                          | -                                            |
| xxiv)                                            | ECT                                                          | See 6.14                                                             | No general requirement                                          | -                                            |
| <b>D - Health , Safety and Environment (HSE)</b> |                                                              |                                                                      |                                                                 |                                              |
| xxv)                                             | Flash point                                                  | IS 1448 ( Part 21 )                                                  | 140 °C, Min ( RDSO Spec.)                                       | 148                                          |
| xxvi)                                            | PCA content                                                  | IP 346                                                               | 3.0 %, Max                                                      | 0.4                                          |
| xxvii)                                           | PCB content                                                  | IS 16082                                                             | Not detectable (< 2 mg/kg)                                      | Not detectable                               |

  
Authorized Signatory